

# Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

melvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

## SUMMARY

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Data scientist with a B.Sc. in Statistics who builds predictive models and tests them properly before anyone relies on them. Ran an uplift study to decide which of 64,000 customers to contact under a fixed budget, built a churn model served on AWS, and chose credit models on calibration. Deploys on GCP Cloud Run and works in Python and SQL.

## SKILLS

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**Machine Learning:** XGBoost, LightGBM, Scikit-learn, PyTorch, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Experiment Design, Statistical Inference, Power Analysis, CUPED, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

**Engineering and MLOps:** Python, SQL, GCP Cloud Run, MLflow, DuckDB, PostgreSQL, Redis, FastAPI, Docker, AWS ECS Fargate, Railway, Prefect, dbt-core, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**AI and LLM Systems:** Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

## PROJECTS

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### Uplift Modelling for Campaign Targeting | [GitHub](#)

*Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions*

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Showed the campaign works (1.25% conversion against 0.57%) but that a non-randomised list overstates the effect by 7.5×, and that a response model mails five times as many customers the campaign harms as an uplift model does.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.

### Telco Customer Churn, End to End on AWS | [GitHub](#)

*Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch*

- Built a churn prediction model with feature engineering and an XGBoost classifier, with experiments tracked in MLflow.
- Served it through a FastAPI /predict endpoint with a Gradio UI, containerised with Docker and deployed on AWS ECS Fargate behind an Application Load Balancer, with CloudWatch logs and GitHub Actions CI/CD.

### Credit Risk Decisioning and Fairness Audit System | [GitHub](#) | [Live App](#)

*Tech: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker*

- Developed a full credit decisioning system producing calibrated default probabilities, cost-optimal approve and decline cutoffs, adverse action reason codes and an append-only audit trail that lets any individual decision be reconstructed months later.
- Benchmarked gradient boosting against a traditional WOE-binned logistic scorecard, prioritising calibration quality (Brier score, reliability diagrams, expected calibration error) over ranking metrics, because expected loss needs true probabilities rather than rankings.

### Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

*Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run*

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.

## EXPERIENCE

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### Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

### Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

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### B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

### Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action